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Battery replacement apparatus and battery replacement method

Abstract

A battery replacement apparatus includes a detection device that detects position and orientation of a battery (first battery), a battery mount where a battery (second battery) is placed, the battery mount being capable of being raised, and an adjuster that adjusts position and orientation of the battery that is placed on the battery mount. The battery mount allows the battery to be attached to a vehicle body from below the vehicle body, by being raised below the vehicle body in a state where the battery is placed on the battery mount. The adjuster adjusts the position and the orientation of the battery placed on the battery mount, based on the position and the orientation of the battery detected by the detection device.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012-192783	12/2011	JP	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This nonprovisional application is based on Japanese Patent Application No. 2022-194354 filed on Dec. 5, 2022 with the Japan Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Field

(2) The present disclosure relates to a battery replacement apparatus and a battery replacement method.

Description of the Background Art

(3) Japanese Patent Laying-Open No. 2012-192783 discloses a battery replacing apparatus for replacing a battery of an electric vehicle. The battery replacing apparatus removes a first battery that is attached to an electric vehicle, and attaches a second battery to the electric vehicle. Each of the first battery removed from the electric vehicle and the second battery is carried on a battery mount portion that is driven below the electric vehicle. A position of the first battery to be removed from the electric vehicle is detected by a laser range finder at a time of removal of the first battery. Moreover, at a time of attachment of the second battery, a position of the battery mount portion is adjusted such that the second battery is moved to the position of the first battery detected by the laser range finder.

SUMMARY

- (4) However, even when the position of the battery mount portion is adjusted as described in Japanese Patent Laying-Open No. 2012-192783, it is sometimes difficult to attach a battery to the electric vehicle due to the position of the battery on the battery mount portion not being constant. Accordingly, a battery replacement apparatus and a battery replacement method by which a battery can be reliably attached to an electric vehicle are desired.
- (5) The present disclosure has been made to solve the problem described above, and is aimed at providing a battery replacement apparatus and a battery replacement method by which a battery can be reliably attached to an electric vehicle.
- (6) A battery replacement apparatus according to a first aspect of the present disclosure is a battery replacement apparatus for replacing a first battery attached to an electric vehicle with a second battery, the battery replacement apparatus including: a detector that detects at least one of position and orientation of the first battery; a battery mount where the second battery is placed, the battery mount being capable of being raised; and an adjuster that adjusts at least one of position and orientation of the second battery that is placed on the battery mount. The battery mount allows the second battery to be attached to a vehicle body of the electric vehicle from below the vehicle body, by being raised below the vehicle body in a state where the second battery is placed on the battery mount. The adjuster adjusts at least one of the position and the orientation of the second battery placed on the battery mount, based on at least one of the position and the orientation of the first battery detected by the detector.
- (7) As described above, with the battery replacement apparatus according to the first aspect of the present disclosure, at least one of the position and the orientation of the second battery placed on the battery mount is adjusted by the adjuster based on at least one of the position and the orientation of the first battery detected by the detector. Accordingly, even when the position and the orientation of the second battery placed on the battery mount are not constant, the second battery may be easily adjusted, by the adjuster, to the position and the orientation suitable for attachment to the electric vehicle. As a result, the second battery may be more reliably attached to the electric vehicle.
- (8) In some embodiments, with the battery replacement apparatus according to the first aspect, the adjuster is driven independently of the battery mount. According to such a configuration, the second battery may be adjusted to the position and the orientation suitable for attachment to the electric vehicle regardless of position and orientation of the battery mount.
- (9) In some embodiments, with the battery replacement apparatus according to the first aspect, the adjuster includes a pre-fixer that pre-fixes a position, in a horizontal direction, of a corner portion of the second battery placed on battery mount, and adjusts at least one of the position and the orientation of the second battery in a state where the position of the corner portion is pre-fixed by the pre-fixer. According to such a configuration, at least one of the position and the orientation of the second battery may be adjusted with reference to the position of the corner portion that is pre-fixed. As a result, control by the adjuster may be simplified.
- (10) In this case, the second battery is rectangular-shaped in plan view. Furthermore, the pre-fixer is L-shaped in plan view. According to such a configuration, movement of the second battery (the corner portion thereof) may be restricted from two directions by the pre-fixer that is L-shaped. As a result, the second battery may be more reliably temporarily fixed.
- (11) A battery replacement method according to a second aspect of the present disclosure is a battery replacement method for replacing a first battery attached to an electric vehicle with a second battery by using a battery replacement apparatus. The battery replacement method includes detecting at least one of position and orientation of the first battery; placing the second battery on a battery mount that is provided in the battery replacement apparatus and that is capable of being raised; adjusting at least one of position and orientation of the second battery that is placed on the battery mount, based on at least one of the position and the orientation of the first battery detected in the detecting; and attaching the second battery to a vehicle body of the electric vehicle from

below the vehicle body, by raising, below the vehicle body, the battery mount on which the second battery is placed, where at least one of the position and the orientation of the second battery is adjusted.

(12) As described above, with the battery replacement method according to the second aspect of the present disclosure, at least one of the position and the orientation of the second battery placed on the battery mount is adjusted by an adjuster based on at least one of the position and the orientation of the first battery detected by a detector. Accordingly, there may be provided a battery replacement method by which the second battery may be more reliably attached to the electric vehicle.

(13) The foregoing and other objects, features, aspects and advantages of the disclosure will become more apparent from the following detailed description of the disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a diagram showing a configuration of a battery replacement apparatus according to an embodiment;
- (2) FIG. 2 is a plan view showing a vehicle stop region of the battery replacement apparatus according to the embodiment;
- (3) FIG. 3 is a diagram showing a configuration of an electric vehicle;
- (4) FIG. 4 is a schematic diagram showing a configuration under floor, in relation to the battery replacement apparatus according to the embodiment;
- (5) FIG. 5 is a perspective view showing a configuration of a battery mount of the battery replacement apparatus according to the embodiment;
- (6) FIG. 6 is a flowchart showing each step by the battery replacement apparatus according to the embodiment;
- (7) FIG. 7 is a bottom view showing the electric vehicle from below;
- (8) FIG. 8A is a schematic diagram of a state where positions of the battery mount and a battery are shifted from each other in a horizontal direction;
- (9) FIG. 8B is a schematic diagram of a state where the shift in the positions of the battery mount and the battery in the horizontal direction is eliminated;
- (10) FIG. 9A is a schematic diagram of a state where a positioning pin is inserted in a pin insertion hole;
- (11) FIG. 9B is a schematic diagram of a state where a locking/unlocking tool is inserted in a tool insertion hole;
- (12) FIG. 9C is a schematic diagram of a state where unlocking is being performed with the locking/unlocking tool;
- (13) FIG. 10A is a schematic diagram of a state where the battery removed from the electric vehicle is moved downward by the battery mount;
- (14) FIG. 10B is a schematic diagram of a state where a lift unit is retracted downward;
- (15) FIG. 10C is a schematic diagram of a state where the battery is housed in a container chamber;
- (16) FIG. 11A is a schematic diagram showing a state where a position of the battery mount is shifted from a position of the battery mount stored in a memory;
- (17) FIG. 11B is a schematic diagram showing a state where the position of the battery mount overlaps the position of the battery mount stored in the memory;
- (18) FIG. 12A is a schematic diagram showing a state where the battery is mounted on the battery mount;
- (19) FIG. 12B is a schematic diagram showing a state where the electric vehicle is raised by the lift unit;

- (20) FIG. 12C is a schematic diagram showing a state where the battery is attached to the electric vehicle;
- (21) FIG. 13A is a schematic diagram showing a state where movement of the battery is restricted by a stopper section;
- (22) FIG. 13B is a schematic diagram showing a state where position and orientation of the battery are adjusted by a movable section;
- (23) FIG. 14 is a schematic diagram showing a radar according to a modification of the embodiment; and
- (24) FIG. 15 is a diagram showing holes provided in the battery of the electric vehicle.

DESCRIPTION OF THE EMBODIMENTS

(25) Hereinafter, an embodiment of the present disclosure will be described in detail with reference to the drawings. Same or corresponding parts in the drawings will be denoted by a same reference sign, and repeated description will be omitted.

(26) (Configuration of Battery Replacement Apparatus)

(27) FIG. 1 is a diagram showing a battery replacement apparatus **100** and an electric vehicle **200** according to a present embodiment. Battery replacement apparatus **100** is an apparatus for replacing a battery **201** attached to electric vehicle **200** with a battery **101**. Battery replacement apparatus **100** includes a battery replacement station **100a** where battery replacement is performed, and a container chamber **100b** where battery **101** is housed. Container chamber **100b** is provided annexed to battery replacement station **100a**. An entrance **102** for allowing electric vehicle **200** to enter and leave is provided in battery replacement station **100a**. Additionally, battery **201** and battery **101** are examples of a “first battery” and a “second battery” of the present disclosure.

(28) Battery **101** housed in container chamber **100b** is moved to a temporary depository **40** provided in an underfloor region S, and is then conveyed to electric vehicle **200**. Additionally, a battery mount **34**, a lift unit **35**, and a conveyor unit **36**, described later, are provided in underfloor region S.

(29) Battery replacement apparatus **100** includes a control device **10**, a detection device **20**, and a drive device **30**. Additionally, detection device **20** is an example of a “detector” of the present disclosure.

(30) Control device **10** includes a processor **11**, a memory **12**, and a communication unit **13**. Memory **12** stores, in addition to programs that are executed by processor **11**, information that is used by the programs (such as maps, mathematical expressions, and various parameters). Although details will be given later, processor **11** controls drive device **30**.

(31) Communication unit **13** includes various communication I/Fs. Processor **11** controls communication unit **13**. Communication unit **13** communicates with a DCM of electric vehicle **200**, and the like. Communication unit **13** and electric vehicle **200** are capable of bidirectional communication. Additionally, communication unit **13** may also communicate with a mobile terminal or the like carried by a user of electric vehicle **200**.

(32) Furthermore, detection device **20** includes a camera **21** and an image processing unit **22**. Details of detection device **20** will be given later.

(33) Furthermore, as shown in FIG. 2, battery replacement apparatus **100** is provided with a vehicle stop region **103**. In a case where an operation instructing start of a battery replacement task is performed by a user through a navigation system, not shown, of electric vehicle **200** in a state where electric vehicle **200** is stopped in vehicle stop region **103**, communication unit **13** receives an instruction signal for starting the battery replacement task from electric vehicle **200**. Processor **11** starts control of the battery replacement task based on reception of the instruction signal by communication unit **13**. Additionally, electric vehicle **200** stops in vehicle stop region **103** in such a way that a front-back direction is an X-direction and a left-right direction is a Y-direction.

(34) Furthermore, referring to FIG. 3, when an ignition power source (not shown) is placed in an off state, electric vehicle **200** places a system main relay (SMR) **204** connecting battery **201** and a

power control unit (PCU) **203** in an off state. Battery **201** is thereby electrically separated from PCU **203** and a motor generator (MG) **205**. When the operation instructing start of the battery replacement task is performed at electric vehicle **200** in this state, battery replacement apparatus **100** starts control of the battery replacement task. Furthermore, due to the operation described above being performed, locking between a vehicle connector **210** and a connector **211** of battery **201** is released. Additionally, after SMR **204** is placed in the off state, power from an auxiliary battery **206** is supplied to accessories such as MG **205** and the DCM via a DC/DC converter **207** and PCU **203**.

(35) Furthermore, drive device **30** includes wheel stoppers **31** (see FIG. 2), a shutter **32** (see FIG. 2), a cleaning unit **33** (see FIG. 2), battery mount **34** (see FIG. 1), lift unit **35** (see FIG. 1), conveyor unit **36** (see FIG. 1), an adjuster **37** (see FIG. 5), and guide sections **38** (see FIG. 13A).

(36) Referring again to FIG. 2, four wheel stoppers **31** are provided in vehicle stop region **103**. Wheel stoppers **31** are provided for four wheels **202** of electric vehicle **200**, respectively.

(37) Wheel stopper **31** includes a pressing member **31a**, a pair of side roller sections **31b**, and a slider section **31c**. Pressing member **31a** is disposed across pair of side roller sections **31b** and slider section **31c**. Pressing member **31a** moves wheel **202** by pressing wheel **202** from outer side (lateral side). Positioning of wheel **202** is thus performed by wheel stopper **31**.

(38) Side roller section **31b** is provided on each of an X1 side and an X2 side of slider section **31c**. Each of pair of side roller sections **31b** is formed from a plurality of rollers, rotation axes of which extend in the X-direction. The plurality of rollers of side roller section **31b** are disposed next to each other in the Y-direction. Pressing member **31a** is moved in the Y-direction when the plurality of rollers of side roller section **31b** are rotated.

(39) Slider section **31c** causes pressing member **31a** placed on wheel stopper **31** to move in the X-direction. Slider section **31c** may be of a conveyor belt type, for example. Additionally, the structure of wheel stopper **31** is not limited to the example described above. For example, side roller sections **31b** or slider section **31c** may be omitted.

(40) Furthermore, processor **11** performs control to clean battery **201** by controlling cleaning unit **33**. Cleaning unit **33** includes two nozzles **33a**, for example. Two nozzles **33a** are provided to sandwich, in the Y-direction, an opening **32a** for retracting battery **201** that is removed from electric vehicle **200**. Nozzle **33a** discharges water toward battery **201** from below battery **201**. Battery **201** is thereby cleaned, and also, a plurality of markers **201d** (see FIG. 7) provided on a bottom surface of battery **201** are cleaned. Additionally, plurality of markers **201d** are provided near different corner portions **201e** (see FIG. 7) of battery **201** having a rectangular shape in plan view (bottom view). Furthermore, the structure of cleaning unit **33** is merely an example, and is not restrictive.

(41) Furthermore, opening **32a** is exposed when shutter **32** is opened. FIG. 2 shows an example where shutter **32** is double-doored, but shutter **32** may instead be single-doored.

(42) As shown in FIG. 4, lift unit **35** lifts or lowers electric vehicle **200** by being lifted or lowered in a state of holding electric vehicle **200** from below. Lift unit **35** is movable in an up-down direction (Z-direction) through opening **32a**.

(43) Lift unit **35** includes a pair of lift bars **35a**. Two protruding portions **35b** protruding on a Z1 side are provided on each of pair of lift bars **35a**. Electric vehicle **200** is supported from below by two protruding portions **35b** of each of pair of lift bars **35a** (that is, by four protruding portions **35b**).

(44) As shown in FIG. 5, battery mount **34** includes two positioning pins **34a**, four locking/unlocking tools **34b**, and a roller section **34c**. Furthermore, camera **21** is mounted (fixed) on battery mount **34**. For example, camera **21** is mounted on an edge portion of battery mount **34**, on a Y2 side. Additionally, camera **21** may instead be provided at a part other than battery mount **34** (such as at lift bar **35a**). Furthermore, camera **21** may be movable relative to battery mount **34**. Additionally, a tapered surface **34d** (see FIG. 9A) is provided at a tip end portion of positioning pin

34a. That is, positioning pin **34a** is tapered toward the Z1 side.

(45) Battery mount **34** is movable in a horizontal direction below electric vehicle **200**. More specifically, battery mount **34** is movable in the X-direction (X1-direction, X2-direction) and the Y-direction (Y1-direction, Y2-direction). Furthermore, battery mount **34** is capable of rotating to change orientation (angle) on an XY plane. Additionally, each of pair of lift bars **35a** may also be movable, as with battery mount **34**.

(46) Referring again to FIG. 1, conveyor unit **36** is capable of conveying a battery (**201**, **101**). More specifically, conveyor unit **36** conveys battery **201** that is removed from electric vehicle **200** and placed on battery mount **34**, to temporary depository **40**. Battery **201** that is placed on battery mount **34** is moved to a Y1 side and placed on conveyor unit **36** due to roller section **34c** (see FIG. 5) of battery mount **34** being rotated in a state where battery mount **34** is lowered to a same height position (a position in the Z-direction) as conveyor unit **36**. Conveyor unit **36** then moves battery **201** to temporary depository **40**. Additionally, conveyor unit **36** may be of conveyor belt type.

(47) Furthermore, conveyor unit **36** moves battery **101** conveyed from container chamber **100b** to temporary depository **40**, to the Y2 side, and places the same on battery mount **34**. At this time, battery **101** is moved to the Y2 side on battery mount **34** due to roller section **34c** of battery mount **34** being rotated in an opposite direction from the case described above.

(48) Referring again to FIG. 5, adjuster **37** includes a stopper section **37a** and a movable section **37b**. Stopper section **37a** is disposed (fixed) on movable section **37b**. Additionally, stopper section **37a** is an example of a “pre-fixer” of the present disclosure.

(49) Stopper section **37a** restricts movement of battery **101** placed on battery mount **34**, to each of the X2 side and the Y2 side. Stopper section **37a** pre-fixes a position, in the horizontal direction, of a corner portion **101a** (see FIG. 13A) of battery **101** placed on battery mount **34**.

(50) In plan view, stopper section **37a** has an L-shape. Additionally, in plan view, battery **101** has a rectangular shape. Accordingly, stopper section **37a** comes into contact with each of a surface **101b** (see FIG. 13A) on the X2 side of battery **101** and a surface **101c** (see FIG. 13A) on the Y2 side.

(51) Furthermore, adjuster **37** is driven independently of battery mount **34**. More specifically, movable section **37b** where stopper section **37a** is disposed is movable in the X-direction (the X1-direction, the X2-direction) and the Y-direction (the Y1-direction, the Y2-direction), independently of battery mount **34**. Furthermore, movable section **37b** is capable of rotating to change orientation (angle) on the XY plane, independently of battery mount **34**.

(52) (Battery Replacement Method)

(53) Next, a battery replacement method that uses battery replacement apparatus **100** will be described with reference to a flowchart (sequence diagram) of FIG. 6.

(54) [Transmission of Vehicle Information, etc.: Electric Vehicle]

(55) First, in step S21, electric vehicle **200** transmits, to communication unit **13** of battery replacement apparatus **100**, information about electric vehicle **200**, information about battery **201**, and information about markers **201d**. For example, when an operation for transmission of each piece of information mentioned above through a navigation system, not shown, of electric vehicle **200** is performed, each piece of information mentioned above is transmitted to communication unit **13**. Electric vehicle **200** transmits each piece of information mentioned above before entering battery replacement apparatus **100**. Additionally, transmission of each piece of information mentioned above may be performed after electric vehicle **200** enters battery replacement apparatus **100**.

(56) [Acquisition of Vehicle Information, etc.: Battery Replacement Apparatus]

(57) Next, in step S1, communication unit **13** of battery replacement apparatus **100** acquires, through communication, the information about electric vehicle **200**, the information about battery **201**, and the information about markers **201d** that are transmitted from electric vehicle **200** in step S21. Each piece of the above-mentioned information that is acquired is stored in memory **12** (see FIG. 1).

(58) More specifically, communication unit **13** acquires information about a size (size class) of electric vehicle **200**. The information about the size of electric vehicle **200** includes information about an overall length and a vehicle width of electric vehicle **200**. Furthermore, the information about the size of electric vehicle **200** may include information about a vehicle height and a ground clearance of electric vehicle **200**.

(59) Furthermore, communication unit **13** acquires information about a size of battery **201**. More specifically, communication unit **13** acquires information about a length L (see FIG. 7) and a width W (see FIG. 7) of battery **201**. Additionally, communication unit **13** may also acquire information about a height H (see FIG. 4) of battery **201**.

(60) Communication unit **13** further acquires position information of battery **201**. More specifically, communication unit **13** acquires position information of battery **201** relative to a vehicle body **200a** of electric vehicle **200**. For example, communication unit **13** acquires information about a distance D1 (see FIG. 7) between a front end portion **200b** (see FIG. 7) of vehicle body **200a** and a front end portion **201a** (see FIG. 7) of battery **201**. Communication unit **13** also acquires information about a distance D2 (see FIG. 7) between a side end portion **200c** (see FIG. 7) of vehicle body **200a** and a side end portion **201b** (see FIG. 7) of battery **201**. Additionally, the position information of battery **201** is not limited to the example mentioned above.

(61) Communication unit **13** may also acquire information about capacity (charge capacity) of battery **201** and state of charge (SOC) of battery **201**.

(62) Moreover, information about marker **201d** (hereinafter “marker information”) includes position information, information about shape, and information about model (such as information about size and color) of each of plurality of markers **201d**. Additionally, the position information of markers **201d** includes information about a position (coordinates) of each of plurality of markers **201d** in a region where battery **201** is disposed, information about positional relationship among plurality of markers **201d**, and the like.

(63) [Transmission of Instruction Signal for Battery Replacement Task: Electric Vehicle]

(64) Next, in step S22, electric vehicle **200** stopped in vehicle stop region **103** transmits the instruction signal for starting the battery replacement task to communication unit **13**. Additionally, after transmitting the instruction signal to communication unit **13**, electric vehicle **200** turns off SMR **204** (see FIG. 3). At this time, communication between electric vehicle **200** and communication unit **13** is maintained by supply of power from auxiliary battery **206** (see FIG. 3).

(65) [Reception of Instruction Signal for Battery Replacement Task: Battery Replacement Apparatus]

(66) Next, in step S2, communication unit **13** receives the instruction signal transmitted by electric vehicle **200** in step S22. Additionally, in step S2, after the instruction signal is received, processor **11** may transmit an instruction message or the like for placing the ignition power source in an off state, to a user of electric vehicle **200** through communication unit **13**.

(67) [Control of Wheel Stopper: Battery Replacement Apparatus]

(68) Next, in step S3, processor **11** adjusts the position of wheel stopper **31** (see FIG. 2) based on the information (vehicle information and battery information) acquired in step S1 through communication unit **13**.

(69) More specifically, processor **11** controls driving of pair of side roller sections **31b** (see FIG. 2) based on information about the vehicle width of electric vehicle **200**. Furthermore, processor **11** controls driving of slider section **31c** (See FIG. 2) based on the overall length of electric vehicle **200**. Moreover, processor **11** controls wheel stopper **31** (side roller sections **31b** and slider section **31c**) based on the position information of battery **201** relative to vehicle body **200a**. Positions of each pressing member **31a** in the X-direction and the Y-direction are thereby adjusted by such control. Additionally, processor **11** may control four wheel stoppers **31** independently of each other.

(70) In this manner, position and orientation of vehicle body **200a** in the horizontal direction are adjusted, and also, position and orientation of battery **201** in the horizontal direction are adjusted.

As a result, battery **201** may be moved to a predetermined position above opening **32a**.

(71) [Cleaning of Marker: Battery Replacement Apparatus]

(72) Next, in step **S4**, processor **11** cleans battery **201** by controlling nozzles **33a** (see FIG. **2**) of cleaning unit **33**. Markers **201d** provided on battery **201** are thereby cleaned. Additionally, cleaning refers not only to removal of stains and foreign substances, but also to removal of snow and ice attached to a lower surface of the vehicle body. Removal of snow and ice is performed using snow-melting agent, hot water, hot air, and the like, for example.

(73) [Horizontal Holding of Vehicle Body: Battery Replacement Apparatus]

(74) Next, in step **S5**, processor **11** places shutter **32** in an open state, and causes lift bars **35a** to be raised in a state where shutter **32** is in the open state. Accordingly, lift bars **35a** pass through opening **32a**, and electric vehicle **200** is lifted by lift bars **35a** (see FIG. **4**). Then, processor **11** raises lift bars **35a** to a predetermined height position, and electric vehicle **200** is thereby held in parallel to the horizontal direction. Additionally, FIG. **4** shows a state where wheels **202** are separated from the ground, but wheels **202** may instead be in contact with the ground.

(75) In step **S5**, processor **11** may adjust a distance by which lift bars **35a** are raised, based on information about the ground clearance of electric vehicle **200** acquired in step **S1**. At this time, lift unit **35** may be controlled such that the ground clearance of electric vehicle **200** becomes constant.

(76) Furthermore, in step **S5**, processor **11** may change a distance (not denoted by a reference sign) between lift bars **35a** based on the vehicle width of electric vehicle **200**. Furthermore, FIG. **5** shows an example where a positional relationship between two protruding portions **35b** of each lift bar **35a** is constant, but the positional relationship between two protruding portions **35b** may instead be adjusted based on the overall length of electric vehicle **200** or the like.

(77) [Detection of Position and Orientation of Battery: Battery Replacement Apparatus]

(78) Next, in step **S6**, detection device **20** (see FIG. **1**) detects position and orientation of battery **201** of electric vehicle **200** held by lift unit **35**.

(79) More specifically, camera **21** acquires an image of three markers **201d** (see FIG. **7**) provided on the bottom surface of battery **201** by capturing battery **201** from below (a Z2 side). Then, image processing unit **22** detects position and orientation of battery **201** based on the image of markers **201d** acquired by camera **21**.

(80) Here, detection device **20** (image processing unit **22**) detects markers **201d** based on the marker information acquired by communication unit **13** in step **S1**. Image processing unit **22** identifies markers **201d** in the image acquired by camera **21**, based on at least one of positional relationship between markers **201d**, information about the shape of markers **201d**, and information about the model of markers **201d**, included in the marker information. For example, image processing unit **22** determines a subject whose shape and model match the information about the shape and model of marker **201d** that is acquired, to be marker **201d**. Furthermore, image processing unit **22** determines a plurality of subjects with a positional relationship that matches the information about the positional relationship between markers **201d** that is acquired, to be markers **201d**.

(81) Then, image processing unit **22** calculates (detects) a region where battery **201** is disposed, based on the positional relationship and the like of three markers **201d**. More specifically, image processing unit **22** detects position and orientation of battery **201** in the horizontal direction. That is, image processing unit **22** detects, in relation to battery **201**, a position in the X-direction, a position in the Y-direction, and orientation (angle) on the XY plane. Furthermore, the position and orientation of battery **201** detected at this time are stored in memory **12**.

(82) Furthermore, image processing unit **22** detects a position (a height position) of battery **201** in the up-down direction (the Z-direction) based on the positional relationship and the like of three markers **201d**.

(83) [Control of Position and Orientation of Battery Mount: Battery Replacement Apparatus]

(84) Next, in step **S7**, processor **11** controls position and orientation of battery mount **34** in the

horizontal direction based on the position and orientation of battery **201** in the horizontal direction detected by image processing unit **22** in step **S6**.

(85) FIG. **8A** is a plan view showing a state where positions of battery **201** (see dashed line) of electric vehicle **200** and battery mount **34** are shifted. When the position and orientation of battery mount **34** are controlled in such a state, a positional shift between battery **201** and the battery mount **34** is eliminated (see FIG. **8B**). Additionally, FIG. **8A** and FIG. **8B** are plan views relative to electric vehicle **200**.

(86) More specifically, the position and orientation of battery mount **34** are controlled such that positioning pins **34a** of battery mount **34** and pin insertion holes **208** provided in electric vehicle **200** overlap in plan view. Locking/unlocking tools **34b** of battery mount **34** and tool insertion holes **201f** provided in electric vehicle **200** (battery **201**) thereby overlap in plan view. Furthermore, the position and orientation of battery mount **34** that are controlled at this time are stored in memory **12**. Additionally, the position and orientation, in the horizontal direction, of at least one of positioning pin **34a** and locking/unlocking tool **34b** may be controlled independently of battery mount **34**.

(87) [Removal of Battery: Battery Replacement Apparatus]

(88) Next, in step **S8**, battery **201** is removed from vehicle body **200a** of electric vehicle **200**. First, processor **11** raises battery mount **34** by a distance that is based on the information about the height position of battery **201** acquired in step **S6**. Accordingly, positioning pins **34a** are inserted into pin insertion holes **208**, and locking/unlocking tools **34b** are inserted into tool insertion holes **201f** (see FIG. **9B**). As a result, battery mount **34** is positioned relative to electric vehicle **200** (battery **201**). Additionally, at this time, positioning pins **34a** are inserted into pin insertion holes **208** before locking/unlocking tools **34b** are inserted into tool insertion holes **201f** (see FIG. **9A**). Furthermore, the height position of battery mount **34** that is raised at this time is stored in memory **12**. Additionally, the distance by which battery mount **34** is raised may be controlled to be constant at all times.

(89) Next, as shown in FIG. **9C**, processor **11** raises locking/unlocking tools **34b** in a state where locking/unlocking tools **34b** are inserted in tool insertion holes **201f**. Then, processor **11** drives (rotates) locking/unlocking tools **34b** inserted in tool insertion holes **201f**. Bolts **201g** inside tool insertion holes **201f** are thereby unlocked. As a result, battery **201** is removed from vehicle body **200a**, and is placed on battery mount **34**. Additionally, a timing of insertion of positioning pins **34a** into pin insertion holes **208** and a timing of insertion of locking/unlocking tools **34b** into tool insertion holes **201f** may be the same.

(90) [Conveying of Battery to Container Chamber: Battery Replacement Apparatus]

(91) Next, in step **S9**, battery **201** removed from vehicle body **200a** in step **S8** is conveyed to container chamber **100b** (see FIG. **1**). First, as shown in FIG. **10A**, processor **11** lowers battery mount **34** on which battery **201** is placed, to a height position of conveyor unit **36** (see FIG. **1**). Next, as shown in FIG. **10B**, processor **11** lowers lift unit **35** (lift bars **35a**) to a position lower than battery mount **34**. Vehicle body **200a** of electric vehicle **200** is thereby placed on the ground without being held by lift bars **35a**. Next, processor **11** drives roller section **34c** (see FIG. **5**) of battery mount **34**. Battery **201** placed on battery mount **34** is thereby moved to the **Y1** side (a conveyor unit **36** side) by roller section **34c**, and is conveyed out from battery mount **34** (see FIG. **10C**). Then, after being conveyed to temporary depository **40** by conveyor unit **36**, battery **201** is housed in container chamber **100b**.

(92) [Control of Position and Orientation of Battery Mount: Battery Replacement Apparatus]

(93) Next, in step **S10**, processor **11** controls the position and orientation of battery mount **34** by moving battery mount **34** based on the position and orientation of battery mount **34** stored in memory **12** in step **S7**. More specifically, in the case (see FIG. **11A**) where there is a shift between the position and orientation of battery mount **34** and the position and orientation of battery mount **34** stored in memory **12** (see dashed line in FIG. **11A**), processor **11** moves and rotates battery

mount **34** in the horizontal direction in such a way that the shift is eliminated (see FIG. **11B**).

(94) [Adjustment of Position and Orientation of Battery on Battery Mount: Battery Replacement Apparatus]

(95) Next, in step **S11**, processor **11** adjusts the position and orientation of battery **101** placed on battery mount **34**. More specifically, processor **11** adjusts the position and orientation of battery **101** by using stopper section **37a** (see FIGS. **5** and **12A**).

(96) First, processor **11** moves battery **101** from container chamber **100b** to temporary depository **40**. Then, processor **11** moves battery **101** from temporary depository **40** to battery mount **34** by controlling conveyor unit **36**. Furthermore, processor **11** moves battery **101** placed on battery mount **34** to the Y2 side by controlling roller section **34c** of battery mount **34**. At this time, battery **101** moved by roller section **34c** stops by coming into contact with stopper section **37a**. Movement of battery **101** to the X2 side and the Y2 is restricted by stopper section **37a** (see FIG. **13A**). A position of corner portion **101a** (see FIG. **13A**) of battery **101** in the horizontal direction is thereby pre-fixed.

(97) Additionally, battery **101** may be moved to the Y2 side by being guided by pair of guide sections **38** (see FIG. **13A**). Furthermore, processor **11** may adjust a distance **D3** (see FIG. **13A**) between pair of guide sections **38** based on information about battery **101** (information about size, type, and the like).

(98) Then, processor **11** adjusts the position and orientation of battery **101** in a state where the position of the corner portion **101a** is pre-fixed by stopper section **37a**. More specifically, processor **11** adjusts the position and orientation of battery **101** placed on battery mount **34**, based on the position and orientation of battery **201** stored in memory **12** in step **S6** (see broken line in FIG. **13A**). More specifically, as shown in FIG. **13B**, processor **11** moves stopper section **37a** (movable section **37b**) in such a way that the position and orientation, in the horizontal direction, of battery **101** placed on battery mount **34** become the same as the position and orientation, in the horizontal direction, of battery **201** stored in memory **12**. Additionally, battery mount **34** is stopped while stopper section **37a** (movable section **37b**) is being moved.

(99) Additionally, position alignment of battery **101** and battery mount **34** relative to each other may be performed by adjuster **37**. In this case, control of the position and orientation of battery mount **34** (control in step **S10**) may be performed after position alignment of battery **101** and battery mount **34** is performed.

(100) [Attachment of Battery: Battery Replacement Apparatus]

(101) Next, in step **S12**, processor **11** performs control of attaching battery **101** to vehicle body **200a**. More specifically, processor **11** raises lift unit **35** (lift bars **35a**) to same height position as lift unit **35** (lift bars **35a**) raised in step **S5**. Vehicle body **200a** is thereby held in parallel with the horizontal direction at the same height position as vehicle body **200a** held by lift unit **35** (lift bars **35a**) in step **S5** (see FIG. **12B**). Additionally, before such control, processor **11** may move lift unit **35** in the horizontal direction such that the position of lift unit **35** in the horizontal direction becomes the same as the position of lift unit **35** in the horizontal direction in step **S5**.

(102) Next, as shown in FIG. **12C**, processor **11** raises battery mount **34** to the height position of battery mount **34** stored in memory **12** in step **S8**. Positioning pins **34a** (see FIG. **5**) are thereby inserted in pin insertion holes **208** (see FIG. **7**). In this state, processor **11** raises locking/unlocking tools **34b**. Locking/unlocking tools **34b** are thereby inserted into tool insertion holes, not shown, of battery **101**. Then, processor **11** drives (rotates) locking/unlocking tools **34b**, and bolts, not shown, in the tool insertion holes are locked. When locking of all the bolts is detected, vehicle connector **210** and a connector, not shown, of battery **101** are locked. As a result, attachment of battery **101** to vehicle body **200a** is completed. Additionally, stopper section **37a** may be retracted (moved away from battery **101**) before battery mount **34** is raised.

(103) [Retraction of Battery Mount and Lift Unit: Battery Replacement Apparatus]

(104) Next, in step **S13**, processor **11** lowers and retracts battery mount **34** and lift unit **35** from

electric vehicle **200**. Then, processor **11** places shutter **32** (see FIG. 2) in a closed state.

(105) [Notification of Battery Replacement Task Completion: Battery Replacement Apparatus]

(106) Next, in step **S14**, processor **11** notifies electric vehicle **200**, through communication unit **13**, of completion of the battery replacement task.

(107) [Reception of Battery Replacement Task Completion Notification: Electric Vehicle]

(108) Then, in step **S23**, electric vehicle **200** receives the notification transmitted from communication unit **13** of battery replacement apparatus **100** in step **S14**. Electric vehicle **200** is thereby placed in a state where the ignition power source can be turned on. Then, the process is ended.

(109) As described above, in the present embodiment, adjuster **37** adjusts the position and orientation of battery **101** placed on battery mount **34**, based on the position and orientation of battery **201** detected by detection device **20**. Accordingly, the position and orientation of battery **101** may be easily adjusted by adjuster **37** regardless of the position and orientation of battery mount **34**. Furthermore, the position and orientation of battery **101** may be adjusted without adjusting battery mount **34**.

(110) In the embodiment described above, an example is described where a position of drive device **30** is adjusted based on information about each of electric vehicle **200** and battery **201**, but the present disclosure is not limited to such an example. The position of drive device **30** may be adjusted based on information about one of electric vehicle **200** and battery **201**.

(111) In the embodiment described above, an example is described where the position and orientation of battery **101** are adjusted by stopper section **37a** being driven independently of battery mount **34**, but the present disclosure is not limited to such an example. Stopper section **37a** may work in conjunction with battery mount **34**.

(112) In the embodiment described above, an example is described where the position and orientation of battery **201** are detected based on an image of markers **201d** captured by camera **21**, but the present disclosure is not limited to such an example. As shown in FIG. 14, the position and orientation of battery **201** may be detected using a radar **121** that emits radio waves (see solid arrows) to battery **201**. More specifically, radar **121** detects a distance to battery **201** based on reflected waves (see dashed arrows) from battery **201**. Additionally, there may be provided a plurality of radars **121**.

(113) In the embodiment described above, an example is described where the position and orientation of battery **201** are detected based on an image of markers **201d** captured by camera **21**, but the present disclosure is not limited to such an example. As shown in FIG. 15, position and orientation of a battery **301** may be detected based on an image of a plurality of holes **301d** provided in battery **301**. Additionally, battery **301** is an example of the “first battery” of the present disclosure.

(114) Although an embodiment of the present disclosure has been described, it is to be understood that the embodiment disclosed herein is by way of illustration and example only and is not to be taken by way of limitation. The scope of the present disclosure is indicated by the appended claims, and is intended to include all modifications within the scope and meaning equivalent to those of the claims.

Claims

1. A battery replacement apparatus for replacing a first battery attached to an electric vehicle with a second battery, the battery replacement apparatus comprising: a detector that detects at least one of position and orientation of the first battery; a battery mount where the second battery is placed, the battery mount being capable of being raised; and an adjuster that adjusts at least one of position and orientation of the second battery that is placed on the battery mount, wherein: the battery mount allows the second battery to be attached to a vehicle body of the electric vehicle from below the

vehicle body, by being raised below the vehicle body in a state where the second battery is placed on the battery mount, the adjuster adjusts at least one of the position and the orientation of the second battery placed on the battery mount, based on at least one of the position and the orientation of the first battery detected by the detector, the adjuster includes a pre-fixer that pre-fixes a position, in a horizontal direction, of a corner portion of the second battery placed on the battery mount, and the adjuster adjusts at least one of the position and the orientation of the second battery in a state where the position of the corner portion is pre-fixed by the pre-fixer.

2. The battery replacement apparatus according to claim 1, wherein the adjuster is driven independently of the battery mount.

3. The battery replacement apparatus according to claim 1, wherein: the second battery is rectangular-shaped in plan view, and the pre-fixer is L-shaped in plan view.

4. A battery replacement method for replacing a first battery attached to an electric vehicle with a second battery by using a battery replacement apparatus, the battery replacement method comprising: detecting, by a detector, at least one of position and orientation of the first battery; placing the second battery on a battery mount that is provided in the battery replacement apparatus and that is capable of being raised; adjusting, by an adjuster, at least one of position and orientation of the second battery that is placed on the battery mount, based on at least one of the position and the orientation of the first battery detected in the detecting; and attaching the second battery to a vehicle body of the electric vehicle from below the vehicle body, by raising, below the vehicle body, the battery mount on which the second battery is placed, where at least one of the position and the orientation of the second battery is adjusted, wherein the adjuster includes a pre-fixer that pre-fixes a position, in a horizontal direction, of a corner portion of the second battery placed on the battery mount, and wherein the adjuster adjusts at least one of the position and the orientation of the second battery in a state where the position of the corner portion is pre-fixed by the pre-fixer.
